

Architecture of 8086 Microprocessor Family

Q. Describe the internal architecture of the 8086 Microprocessor. Explain the functions of the Bus Interface Unit (BIU), Execution Unit (EU), Registers, and Memory Segmentation.

****Definition to Remember****

The ****In**
internal
****Execu**

1. Bus Interface Unit (BIU)

Handles all data and address transfers on the system buses. The EU relies entirely on the BIU to fetch data.

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2. Execution Unit (EU)

Responsible for decoding and executing instructions.

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operatio

3. Registers of 8086 (16-bit)

| Category | Registers |

| :--- | :-

4. Memory Segmentation

The 8086 has 16-bit registers but needs to access 20-bit addresses (1 MB). Memory is logically divided into segments of 64 KB each.

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****Must-Write Points (for 10 marks)****

1. 8086

*****Quick Recall*****

`8086 ->
Register